Kasin Ito-Sato

Detailed Design Report

CDR Revisions:

Personal:
My design takes the input from the fan on the floor four feet away by pushing a ball into a cup. This cup is attached to a pulley that when weighed down on one side, will release another ball at the top of the 48” by 48” plywood sheet. This ball will roll down and affect another similar pulley system. This pulley will release another ball hitting a block starting a domino effect. The domino effect will then release another ball that will roll along a slanted ramp and knock off hooks holding pieces of paper to reveal the letters “L”, “M”, and “U”. Throughout my design I prioritized functionality the most when creating my design. I wanted to include parts that would consistently work and did not rely on chance. The pulleys and ramps will most likely work consistently the only part of my design that may have a chance of malfunctioning is the domino portion they either fall off or do not fall.

For Kasin’s design, we chose a score of eight for functionality as the concept should work effectively but some features have a chance of malfunctioning. The features that may malfunction are most likely the domino aspect as they may fall off of the design or not be placed a proper distance apart and not fall over. We then chose a score of seven for tempo because there are a good number of moving parts but also areas that could be considered slow. The areas that may be slow are the domino and LMU letter reveal as the dominoes may not fall fast and the letter reveal depends on the velocity of the ball down the ramp. The complexity of the design, however, will dry interest and anticipation for the next event to occur. We then gave a score of eight for surprise because of the pulleys and the reveal of the LMU letters. The pulleys reveal a ball which adds some surprise but the reveal of the LMU letters could be improved and made more interesting. Finally, for comedy, we gave a score of nine because of the many moving parts that add to the overall enjoyment of the system.

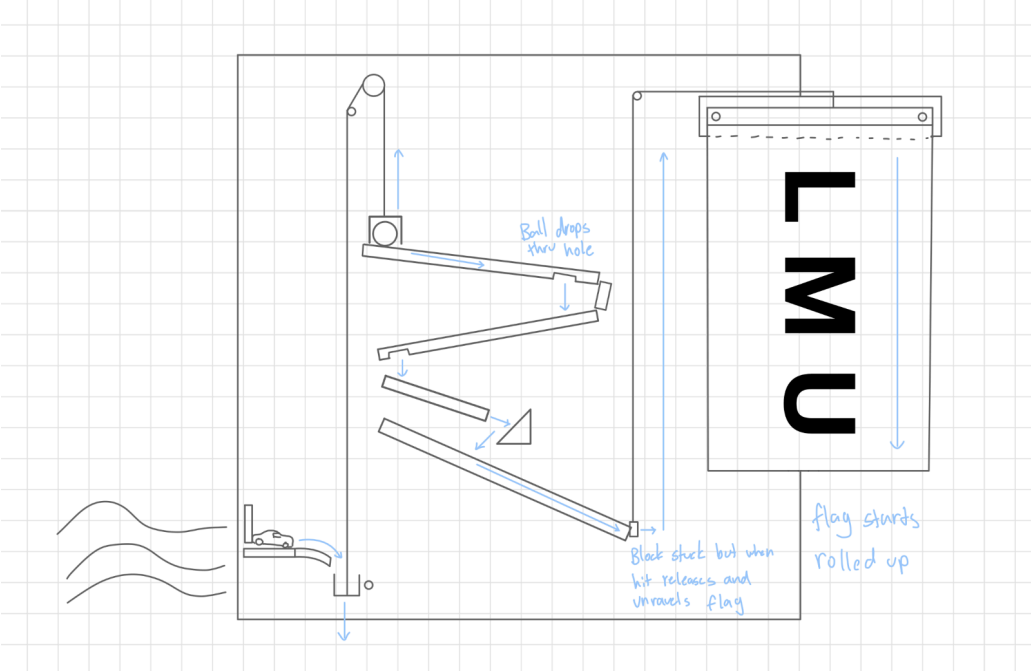
Group:
We chose to go with a design like Kasin’s design. All our designs were somewhat similar, but Kasin’s seemed to be a combination of our other two designs and had a generally good scoring in each of our categories. We are not tied down to following this design exactly and want to continue to workshop what could add more interest and value to our design. One of the places we think we should think further about is the reveal of the LMU letters. This wasn’t the deciding factor, but he also received the

highest score which helped influence us to pick his design. We liked his tempo and overall complexity, shown in our scores, better since it was just laid out better and an interesting design, we think would keep people entertained. The main difference between our designs was the section between the input and output. Issac and Kasin had a similar idea to use a pulley as the input as it would allow us to use gravity to our advantage and Cooper agreed with this idea. Throughout the middle section of our design, we all used ramps but in Kasin's he also included a domino section that may add some interest. Issac's design also had a slightly longer ramp section than Cooper's design.

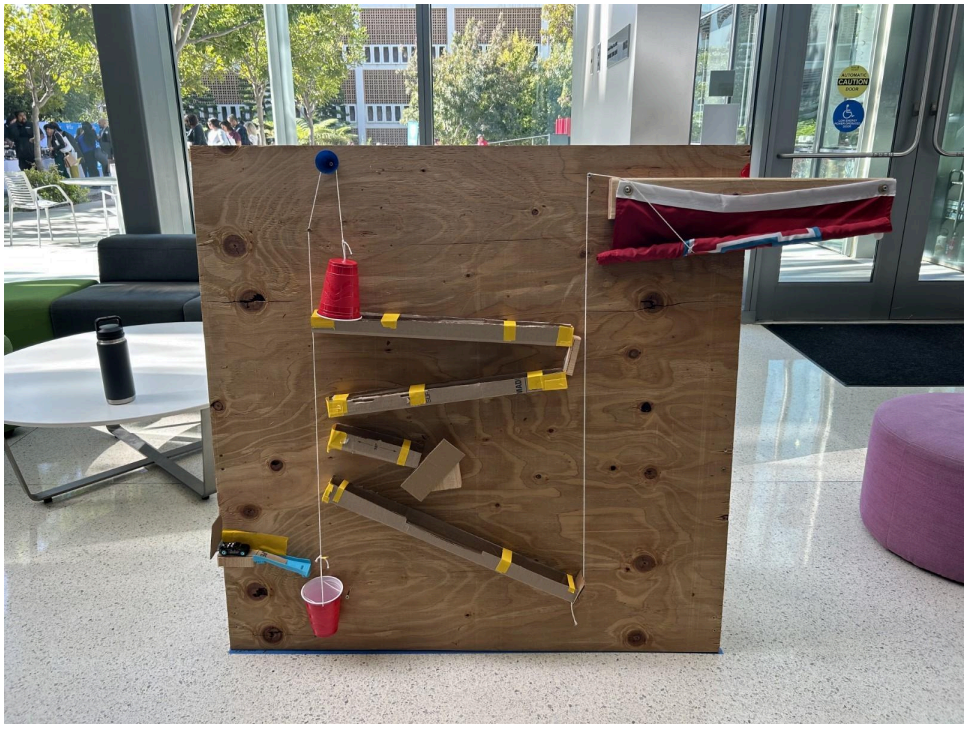
DDR:

Design Implementation:

Final Sketch:



Actual Design:



Our design starts with an input from a box fan four feet away and the wind from this fan is caught by a car with a large rectangular piece of cardboard attached to its back. This car then falls into a cup which activates a pulley that releases a large metal ball down a set of ramps. The ramps go back and forth down our design and have holes where the ball falls through or a backboard that the ball bounces off of into the next ramp. These ramps then lead to the bottom of the board and the ball ends up hitting a piece of cardboard attached to a string that is lightly taped to the last ramp. This string is holding the rolled-up LMU flag and when the cardboard piece is hit, it gets dislodged from the ramp and the flag is released revealing the LMU letters one by one. The flag has a pole attached to the bottom which allows it to be heavy and roll down nicely.

Design Evaluation:

Our design was successful in completing the task of using the box fan as input and having the LMU letters be revealed one by one at the end. Our input using the pulley and car activation was able to work consistently as predicted in our analysis. One of the first challenges we faced was ensuring the ball was kept on the ramps. There were times when the ball would hit an edge or roll weirdly and fly off the board, but we combatted this by adding extra borders and pieces of wood to ensure it would stay on track. Our largest challenge was figuring out a consistent way to have the flag roll down. We tried many different methods like adding screws as contact points, using pipe cleaners so the string moved smoother, and even using the rim of a red solo cup in hopes the piece of cardboard could fit through it and allow the flag to fully roll down. We

eventually found a consistent method that only needed one screw placed strategically above the end of the last ramp. This allowed the cardboard piece to not get stuck on anything as well and only having one contact point meant the string would move smoothly and release the flag effectively.